Project Documentation

ArenaX Gaming Website · DailyBrief News Portal · Travel Booking System

HTML5 · CSS3 · Vanilla JavaScript

1. Introduction

This document describes three single-page websites built with only HTML, CSS and JavaScript. Each project is one self-contained **.html** file that runs by opening it in a browser. Together they demonstrate layout design, responsive CSS, DOM manipulation, event handling and form validation.

Project	File	Purpose	Uses JavaScript?
ArenaX	Gamingwebsite_.html	Esports community landing page	No
DailyBrief	Hussain_News.html	News homepage with search and newsletter	Yes (about 40 lines)
Travel Booking System	train_booking_.html	Bus/train search, seat selection, fare, confirmation	Yes (about 170 lines)

2. How to Run

- Place the HTML files in a folder and open any of them in Chrome, Edge, Firefox or Safari.
- No server, database, or installation is required.
- DailyBrief loads images from Unsplash, so an internet connection is needed to see photos.

3. Common Techniques

Technique	Where used	Why
CSS Grid with auto-fit / minmax	All three	Cards re-flow automatically on any screen width
Flexbox	Travel header and tabs	Align logo, menu and tab buttons
CSS variables (:root)	Travel	Change a colour once, updates everywhere
Media queries	All three	Phone-friendly layouts (≤ 600 px / ≤ 480 px)
Hover and transition effects	Gaming, News	Visual feedback on cards and buttons
Anchor navigation (#id)	All three	In-page menu without extra pages
Template strings and innerHTML	News, Travel	Generate cards and rows from data

4. ArenaX – Gaming & Esports Website

4.1 Overview

A dark-themed landing page for a fictional esports community. It is purely static: structure in HTML, presentation in CSS, no scripting.

4.2 Page Structure

Section (id)	Content
header	Logo "ARENAX" and navigation links (Home, Games, Tournaments, Players, Leaderboard, News)
#home (hero)	Headline "PLAY HARD. RISE HIGHER.", short tagline and an "Explore Tournaments" button
#games	Four cards: Valorant, League of Legends, Counter-Strike 2, Clash of Clans, each with a genre tag
#tournaments	ArenaX Valorant Open (26 Oct 2026, $5,000 prize), Summoner's Clash (29 Oct 2026), Counter-Strike Night Cup (2 Nov 2026)
#players	Player cards: Nyx (Valorant, Duelist), Razor (CS2, AWPer, rating 1.36), Lux (LoL, Mid Lane, Top 0.5%)
#leaderboard	Table of five teams with rank, game, wins and points (Nova Core leads with 2840)
#news	Three news cards in a grid
footer	Copyright and tagline

4.3 Design

Item	Value
Background	#0b0e17 (near black)
Accent colour	#a8ff3e (neon green)
Card colour / border	#171d2b / #293148
Hero	Linear gradient #191f35 → #35205c
Font	Arial, sans-serif

4.4 Key CSS Behaviour

- **.games / .news**: `grid-template-columns: repeat(auto-fit, minmax(220px, 1fr))` creates a fluid multi-column layout.
- **.card:hover**: moves up 5 px, border turns green and a soft green glow appears (0.3 s transition).
- **.button:hover**: turns white and lifts 3 px.
- **Table**: green header row, row hover highlight, points column in bold green.
- **@media (max-width: 600px)**: smaller headline, tighter section padding, smaller table text and wrapped nav links.

4.5 Limitations

- All data (teams, players, dates) is hard-coded.

- "Registration Open" is only a label; there is no sign-up form.
- Nav links jump within the page only.

5. DailyBrief – Online News Portal

5.1 Overview

A light news homepage with six category cards, a client-side search and a demo newsletter form. Navy (#14213d) and orange (#fca311) form the palette.

5.2 Page Structure

Section	Content
header	Site title "DailyBrief" and tagline
nav	Home, National, International, Sports, Technology (anchors to cards)
.search-box	Text input plus *Search* and *Show All* buttons
main .container	"Latest News" grid: National, International, Sports, Technology, Business, Education
.newsletter	Email input and Subscribe button, message paragraph
footer	Copyright and "Created by Students" line

5.3 JavaScript Functions

Function / handler	What it does
searchNews()	Reads the input, lowercases and trims it, then loops over every **.news-card**. If the card's `innerText` (category, title and description) contains the text, it is shown; otherwise `display:none`.
showAllNews()	Shows every card again and clears the search box.
keydown listener	Pressing **Enter** in the search box runs `searchNews()`.
submit listener (newsletter)	Calls `preventDefault()`, shows "Thank you for subscribing!" and clears the email field. No data is sent anywhere.

5.4 Responsive Behaviour

The grid uses `minmax(250px, 1fr)`, giving four columns on a wide desktop and one on a phone. Below 600 px the nav links become inline-blocks and the search input widens to 90%.

5.5 Limitations and Notes

- Images are fetched from Unsplash; they will not show offline.
- "Read More" links only scroll to an anchor; there are no article pages.
- An empty search matches everything, so all cards remain visible.
- Search is case-insensitive but only substring-based (no ranking or typo tolerance).

6. Travel Booking System (Bus & Train)

6.1 Overview

A student demo that walks the user through a complete booking: search, choose a service, pick seats, enter passengers and view a confirmation. State is kept in JavaScript variables only (no backend), and the interface is revealed step by step by toggling a **.hidden** class.

6.2 User Flow

Step	Action	Result
1	Choose Bus or Train tab (or nav link)	`setMode()` updates tabs, button label and hides old results
2	Fill source, destination, date, passengers	Inputs validated on Search
3	Click Search	Three service cards appear (name, times, duration, price)
4	Click "Select & choose seats"	Seat map, passenger fields and fare appear; 5 random seats are shown as booked
5	Click seats	Seats toggle green; cannot exceed passenger count
6	Click Book Ticket	Validated, then confirmation table is shown

6.3 Sample Data

Mode	Service	Departs	Arrives	Duration	Price
Bus	Express Star	06:00 AM	02:00 PM	8h 00m	■650
Bus	Night Rider	09:30 PM	06:30 AM	9h 00m	■800
Bus	City Link AC	01:00 PM	08:30 PM	7h 30m	■720
Train	Coastal Express	07:15 AM	03:45 PM	8h 30m	■420
Train	Southern Superfast	05:40 PM	05:10 AM	11h 30m	■560
Train	Intercity Special	10:00 AM	06:00 PM	8h 00m	■380

6.4 JavaScript Functions

Function	Purpose
setMode(newMode)	Switches between "Bus" and "Train"; toggles the active tab, updates the search button label, hides services, seats and confirmation.
searchServices()	Validates the form (all fields filled, source ≠ destination), reads passenger count, builds one card per service with a loop and shows the list.
chooseService(index)	Stores the chosen service, highlights the card, then calls *makeBookedSeats*, *drawSeats*, *drawPassengerFields* and *updateFare*.
makeBookedSeats()	Picks 5 unique random seat numbers (1–20) as already booked.
drawSeats()	Creates 20 seat buttons. Bus layout is 2 + aisle + 2; Train layout is 3 + aisle + 2. Booked seats are disabled and grey.
toggleSeat(no, btn)	Selects or unselects a seat; shows an error if more seats than passengers are chosen.
drawPassengerFields()	Creates one name + age row per passenger.
updateFare()	Total fare = price per ticket × number of passengers.
bookTicket()	Checks seat count and each passenger's name and age (1–120), then fills the confirmation table (service, route, date, timing, passengers, seats, total fare).

6.5 Validation Rules

- Source, destination and date are required.
- Source and destination must differ (case-insensitive).
- Date picker minimum is today, so past dates are blocked.
- Exactly as many seats as passengers must be selected.
- Every passenger needs a name and an age from 1 to 120.

6.6 Design

CSS variable	Colour	Used for
--dark	#12343b	Header, headings, seat outline
--accent	#f2b705	Buttons, highlighted service
--green	#2e9e6b	Selected seat, confirmation
--grey	#c9d3d5	Booked seat
--red	#c0392b	Error messages
--light	#f3f7f6	Page background

6.7 Known Issues and Suggested Fixes

Issue	Suggested fix
The file ends with `</html` (missing >)	Add the closing >.
Changing the passenger dropdown after searching does not refresh seats or fare	Re-run the search, or add a change listener on the dropdown.

Issue	Suggested fix
Min date uses `toISOString()` (UTC), which can be one day off near midnight in some time zones	Build the date string from local year/month/day.
Source and destination do not change prices or timings	Add route-based data or an API.
Booked seats are random on every selection	Store bookings per service and date.

7. Testing Summary

Each page was opened in headless Chromium and the main interactions were exercised. All behaved as designed.

Project	Test	Result
ArenaX	Desktop and mobile render; sections and table visible	Pass
DailyBrief	Search "tech" filters cards; Show All restores; newsletter shows thank-you message	Pass
DailyBrief	Images load	Not testable offline (external Unsplash URLs)
Travel	Empty search shows validation error	Pass
Travel	Hyderabad → Chennai, 2 passengers, Train: services listed	Pass
Travel	Select 2 seats, enter passengers: fare = ■420 × 2 = ■840	Pass
Travel	Book Ticket shows confirmation	Pass
Travel	Mobile layout (390 px)	Pass

8. Conclusion

The three projects show how far plain HTML, CSS and JavaScript can go: a static, styled landing page; an interactive page with search and forms; and a multi-step booking application with validation and dynamic content. Natural next steps are connecting real data through an API, adding user accounts, and running an accessibility review.